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Automated Resume Screening Using Machine Learning: An Ensemble-Based Approach for Efficient Candidate Selection

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Abstract: Large scale recruitment is time-consuming and results in inefficiencies where manual screening of resumes is used. This study proposed a novel Machine Learning (ML) based ensemble framework for resume classification to improve the process of recruitment. The research has a two-phase experimental design. First, several ML classifiers such as Logistic Regression, Decision Trees, Naive Bayes and Random Forest have been benchmarked to determine baseline performance. Secondly on the basis of these results, the best performing models were merged in a strategic manner to form ensemble framework using soft-voting mechanism. The proposed ensemble framework performed significantly better than standalone classifiers on a standard benchmark data set with an accuracy of 98.44% (5-fold CV mean: 97.47% ±1.00%). Paired T-tests ($p<0.05$) were used to ensure the statistical significance, which confirmed the reliability of the performance gains. In order to resolve the robustness gap in literature, the framework was tested on a new curated expert-labeled heterogeneous dataset (N=598) with a Zero-Shot testing protocol. The ensemble showed outstanding stability with a performance decrease of just 1.10 percent, and an F1-score of 0.99. Additionally, an interpretability audit was used to confirm algorithmic fairness, which guarantees the selection based on merit. Finally, a full functional web-based prototype was created to illustrate practical viability in real-time resume classification and recommendation. The results prove that the framework is not only highly accurate and statistically reliable, but also practically implementable for real-world talent acquisition.

Keywords: Classification, Ensemble Model, Machine Learning, Resume Screening, Recommendations

1. Introduction

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In the current landscape of human recruitment and the globalized digital labor market, talent acquisition has transitioned from a localized administrative function into a data-centric strategic imperative [1]. The exponential proliferation of professional networking platforms and digital application portals has resulted in an overwhelming influx of unstructured candidate data, fundamentally shifting the paradigm [2]. To preserve organizational agility and competitive advantage, modern enterprises are increasingly adopting Artificial Intelligence (AI) and ML to streamline the recruitment process [3,4]. The synergy between Natural Language Processing (NLP) and supervised learning frameworks allows for the automated extraction of high-dimensional features from diverse resume corpora, facilitating a more objective, rapid, and data-driven candidate selection process that aligns with global professional competencies [5]. Consequently, the development of sophisticated, automated screening architectures is no longer merely a logistical convenience but a vital necessity for organizations managing large-scale recruitment while striving to maintain the highest standards of talent quality and operational efficiency.

However, the rapid adoption of these technologies has outpaced the development of robust and reliable evaluation frameworks, leading to significant operational hurdles in practical recruitment workflows. Timely recruitment and meritocratic judgment frequently suffer due to the inefficiencies of legacy manual screening, which is inherently susceptible to cognitive heuristics and human biases [13]. A primary technical bottleneck arises from the immense structural heterogeneity of digital resumes; HR departments are confronted with a vast array of non-standardized formats, multi-columnar layouts, and inconsistent linguistic styles that often baffle traditional parsing algorithms. Despite the success of supervised learning in controlled, high-resource settings, these models often exhibit a sharp decline in performance when confronted with the noisy and varied data distributions typical of a globalized workforce. This has created a persistent Trust Deficit in AI recruitment tools, necessitating the engineering of screening systems that are not only accurate on paper but are also robust, efficient, and transparent enough for industrial deployment.

A critical synthesis of the existing literature reveals several fundamental research gaps that remain unresolved. Foremost is the Robustness Gap; the majority of contemporary research (e.g., [5], [7]) focuses almost exclusively on maximizing accuracy within static, pre-cleaned benchmark datasets like Kaggle. There is a profound lack of empirical evidence regarding how these models generalize to unseen data from heterogeneous sources, leaving a vital question of practical resilience open. Second, an Efficiency Paradox is evident in the recent pivot toward heavy DL and transformer-based architectures. Despite their immense computational cost and latency, literature [10, 18] suggests these models often fail to provide a statistically significant performance gain over optimized classical ensembles in text-heavy resume classification. Third, a Layout Rigidity Gap persists where

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existing parsers fail to handle non-standardized PDF structures, often losing critical semantic information during the data extraction phase. Finally, an Operational Disconnect exists, as most academic contributions terminate at the model-validation stage, failing to provide an end-to-end functional prototype that integrates classification, ranked recommendation, and interpretability audits into a single HR-ready workflow. Driven by these challenges, the motivations of this research are:

1. The digital job applications have increased exponentially and the manual screening has become the most important bottleneck to the modern HR departments. Although automation exist, a trade-off between speed and classification accuracy persists. The rationale behind this study is the creation of a highly accurate system that will be able to handle huge amounts of disordered resumes in real-time, eliminating a huge amount of cognitive load on the recruiters without compromising the highest degree of accuracy in filtering the candidates.
2. Conventional screening of resumes is usually vulnerable to human biases, which results in biased recruitment. One of the main reasons why this research is needed is the shift towards fair recruitment process. This study seeks to address the issue of subjective bias by using ML to filter down to semantic professional indicators (skills, experience, and certifications) to make sure that the selection of candidates is based on objectively founded technical competencies and not demographic factors.
3. The majority of current automated screening systems work remarkably well with clean and standardized data sets but do not work well when subjected to the variety of formatting, inconsistent language styles, and structures of real world resumes. This study is motivated by the critical need for a novel screening solution. This study aims to develop a novel framework to ensure that performance is maintained over a wide range of data distributions and unknown formats, by proposing a multi-logic ensemble framework.

Based on these motivations following are research questions:

RQ1: To what extent can a hybrid ensemble framework improve upon the benchmark accuracy of high-performing individual classifiers in the context of automated resume screening?
RQ2: Beyond standard accuracy metrics, how does the real-world robustness of a hybrid ensemble framework compare to that of standalone classifiers when evaluated on a diverse, unseen, and heterogeneous dataset?
To address the identified research questions, this paper presents the following contributions.

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1. Instead of using one classifier this study proposed an Ensemble Framework that combines geometric (Support Vector Machine), probabilistic (Naïve Bayes), and tree-based (Random Forest) logic in a strategic way. Through the use of a Soft-Voting mechanism that has been specifically adjusted to solve lexical ambiguities and high-overlap groups in resume text, the proposed framework achieve a better accuracy than single models.
2. The study represents a major methodological change, as it goes beyond the conventional single dataset assessments. We offer a evaluation stage by applying a Zero-Shot testing protocol to a newly curated, heterogeneous dataset (N=598). This evaluation empirically confirms the stability and generalizability of framework in disparate data distributions in the real world, practically closing the gap between accuracy and practical implementation.
3. This paper provides a strict post-hoc interpretability and algorithmic fairness framework through the conduction of an empirical audit of feature importance. This contribution goes beyond the black-box classification by showing that the decision-making logic of the model is entirely based on professional competencies and merit-based keywords, which further offers verifiable evidence of the capability of the system to reduce the subjective human bias in talent acquisition.
4. The research shows that it is practically viable by creating a complete web-based system for screening resumes. The prototype is a demonstration of how proposed framework can be incorporated into HR processes to streamline the efficiency of talent acquisition without sacrificing the objectivity and merit-based selection of candidates, thus offering a scalable solution to the problem of subjective human bias.

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The rest of this paper is structured as follows. Section 2 is the detailed review of the related work to define the context of the research and define the gaps that exist. Section 3 outlines the proposed methodology, including the process of data collection and preprocessing, as well as the structure of the ensemble framework. Finally, Section 4 presents and analyzes the experimental results, while Section 5 concludes the paper by summarizing the key findings and their implications.

2. Related work

As the world of recruitment continues to evolve, automated screening of resumes has become an essential tool for managing the massive volume of applications received and to overcome the subjectivity of human decision making. During this initial phase, the researchers were concerned with demonstrating the feasibility of the traditional algorithms on text-based applications. Daryani et al. [4] were the first to introduce a system that

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matches a resume to a job description based on a Vector Space Model and cosine similarity. This employed mathematical models to solve document alignment problems but the main drawback of this work was its limited functionality, it was only used as a matching tool and did not classify the candidates into various professional fields. Later, Roy et al. [1] proposed a more systematic two-phase system which implements a Linear SVM to achieve an accuracy of 78.53%. A key limitation of this study was that the reliability was moderate, with an accuracy rate of less than 80% which would not meet the criteria for high-stakes corporate hiring where a misclassification might cause considerable financial or operational loss.

This performance gap inspired a surge of studies aimed at achieving maximum classification accuracy with benchmarking various classical ML classifiers. Kinge et al. [5] and Fareed et al. [6] achieved an accuracy of 98.96% in both cases, on standardized Kaggle datasets, using a KNN and a multi-classifier approach, respectively. This approach peaked with Ali et al. [7] which used an optimized Linear SVM to achieve an almost perfect accuracy of 99.6%. To complement this period, AR et al. [18] investigated several models, including SVM and Naive Bayes, and determined that XGBoost was a suitable model for a user-friendly web application, with an AUC-ROC of 0.89. The key drawback of this approach is that individual model lack collective intelligence as ensembles and its accuracy is not as high as benchmarked standards under diverse professional categories. Even with high numbers, a Benchmark Bias became the big shortcoming of this period, as these studies took place in controlled setting with pre-cleaned data. None of these models were evaluated on unseen test sets or even tested against noisy data from any other sources like LinkedIn or different PDF formats, which makes their practical resilience and applicability to real data questionable.

However, it became apparent that the raw classification was insufficient, and attention turned to NLP and more sophisticated feature engineering to gain a deeper understanding of the semantics. Sharma et al. [8] tried to innovate this space by proposing an "Elite Bag-of-Words" system that aims to identify high-value business keywords. Interestingly, this approach achieved an accuracy of only 62.60%, demonstrating a Complexity Trap, which occurs when complex feature engineering underperforms simpler ones such as TF-IDF. In the latest developments, Deshmukh and Raut [19] used NLP with BERT for candidate ranking, computing similarity indices at one resume per second. The study, however, faces a major Data Scalability Gap, as they only tested 200 resumes; and the similarity threshold of 0.3 for their model suggests that it may not be accurate enough for use in competitive hiring situations. Similarly, Najjar et al. [11] have used Word2Vec for semantic ranking and have achieved 96% accuracy. In addressing the ethical issues of recruitment, Pulavarthi et al. [20] suggested a scalable approach through NLP and ML, aiming for a more fair process by mitigating demographic bias. One of the key weaknesses of the algorithm presented here is the Fairness-Precision Trade-off, which is that reducing bias is a necessary condition but the de-emphasis of certain credentials can also mask candidate quality and leave a room for a system that

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is both technically precise and algorithmically just. In addition, Pampouktsi et al. [12] created a model for the recruitment process in the public sector which achieved an accuracy rate of 96.93% but the disadvantage of such specific systems is that they do not perform well when implemented in other organizational settings.

In the end, due to the intrinsic constraints of manual feature engineering, a DL and Transformer structures were used to automatically extract patterns of a higher complexity. Shen et al. [9] compared various neural models for resume parsing and determined that a CRNN can achieve 96% accuracy. Karuppiah et al. [21] further pushed this trend by using SBERT and FAISS to perform semantic matching on large-scale intelligent platforms with a top-3 accuracy of 99.21%. Even with this level of accuracy, the research finds an Operational Latency Gap: the computational requirements of transformer ensembles and generative modules can result in scalability challenges in high volume, real-time industrial applications. The Complexity-Performance Paradox was also brought to light by Kuppala et al. [10] who applied a sophisticated CNN+GRU ensemble. Surprisingly, their model only achieved an accuracy of 88% compared to 99.6% of simple Linear SVMs on the same data. More recently, Abhishek et al. [22] designed an AI-powered analyzer that uses Pyresparser and NLTK to provide instant recommendations. The major disadvantage with this tool is the Layout Rigidity, which means that it will only work with specific parsing libraries; it will not work with non-standardized or multi-columnar PDF layouts which are typical in the modern global job landscape. The key discovery is that sometimes deep learning, heavy parsing libraries are not the best approach, as they can be less efficient and less accurate than classical approaches.

More recent discussion has moved beyond ethical issues and to more sophisticated ensemble concepts. However, Harris [13] illustrated the critical Fairness Gap, showing that AI screeners can perpetuate and escalate human biases, including ageism. In the recruitment area, Al-Quhfa et al. [14] showed that Random Forest gave an accuracy of 92.8%. However, one of the big challenges was always that same: the absence of external, non-benchmark testing.

Table 1: Comprehensive Literature Survey and Research Gap Mapping

Reference	**Methodology**	**Major Contribution**	**Reported Performance**	**Critical Limitation / Identified Gap**
Daryani et al. [4]	Vector Space Model	Proved viability of resume-to-JD matching.	N/A	Functional Gap: Strictly a matching tool; cannot perform domain classification.

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Roy et al. [1]	Linear SVM	Developed a two-phase classification/ranking system.	78.53%	Reliability Gap: Performance is too low for sensitive hiring decisions.
Kinge et al. [5]	KNN	High-accuracy benchmarking on Kaggle data.	98.96%	Benchmark Bias: Confined to cleaned, structured data only.
Fareed et al. [6]	Multi-Classifier	Extensive comparison of standalone ML models.	98.96%	Robustness Gap: No empirical evidence of performance on noisy, unseen data.
Ali et al. [7]	Optimized SVM	Peak reported accuracy for benchmark datasets.	99.6%	Overfitting Risk: High score on seen data fails to ensure real-world resilience.
Sharma et al. [8]	Elite BoW	Novel keyword extraction methodology.	62.6%	Complexity Trap: Performed worse than basic algorithms like TF-IDF.
Najjar et al. [11]	Word2Vec	Semantic ranking for candidate selection.	96.0%	Scalability Gap: Limited to a tiny, domain-specific (IT) dataset of 101 resumes.
Pampoukt si [12]	ML Models	Specialized screening for public sector recruitment.	96.93%	Generalization Gap: Niche models lack versatility across different industries.
Shen et al. [9]	CRNN	Automated parsing and feature extraction.	96.0%	Operational Gap: High computational cost; rigid model

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				architecture.
Kuppala et al. [10]	CNN + GRU	Hybrid neural ensemble for resume data.	88.0% Acc	Efficiency Paradox: Significantly lower accuracy than classical ML models.
Harris [13]	Demographic Audit	Identified age bias inheritance in AI screeners.	N/A	Fairness Gap: Focuses on ethics but lacks a robust technical classification solution.
Al-Quhfa [14]	Random Forest	Balanced performance for talent recruitment.	92.8% Acc	Validation Gap: Effective model but lacks "Zero-Shot" testing on external data.
AR et al. [18]	XGBoost	ML-based web application.	0.89 AUC	Accuracy Gap: Single model performance is lower than ensembles.
Deshmukh [19]	BERT & NLP	Keyword extraction/ranking.	1 res/sec	Data Gap: Tested on only 200 resumes; low similarity threshold.
Pulavarthi [20]	NLP & ML	Addressing efficiency/fairness.	N/A	Fairness Trade-off: Bias removal may impact technical precision.
Karuppiah [21]	SBERT & FAISS	AI Talent Acquisition Platform.	99.21%	Operational Gap: High computational cost and latency of transformers.
Abhishek [22]	Pyresparser/NLTK	CV Analysis/Recommendat	N/A	Rigidity Gap: Fails on non-standard

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		ion.		layouts and multi-column PDFs.

As illustrated in Table 1, the synthesis of existing literature reveals four fundamental research gaps that the present study aims to address. First is the Robustness Gap; despite achieving near-perfect accuracy on benchmark datasets, there is a lack of empirical evidence regarding model performance on unseen, heterogeneous data distributions. Second is the Efficiency Paradox [10, 23], where increased computational complexity has failed to yield superior accuracy or real-time scalability compared to optimized classical models. Third is the Data Scalability and Rigidity Gap [21, 24], where models are either tested on small samples or fail to parse non-standardized layouts. Finally, there is an Operational Gap, as most academic works lack an end-to-end functional prototype.

3. Methodology

To address the gap mention in above literature this research follows the Design Science Research (DSR) Methodology, which consists of three primary phases: Problem Identification, Prototype Design and Development and Prototype Demonstration and Evaluation. Each phase is further divided into specific activities that contribute to the systematic development of the proposed automated resume screening framework.

1) Phase 1: Problem Identification is concerned with the research gap and outlining the study objectives. The activities within this phase include Literature Review and Rationale Understanding, Identification of Research Question and Identification of Research Objective. These points have been mentioned already in the Introduction and Literature review section.
2) Phase 2: Prototype Design and Development entails the technical implementation of the system. This stage includes three major activities: Activity 1 Data Collection and Preprocessing, in which the resume data is collected and processed using the NLP methods of text cleaning and extracting features. Activity 2: Proposed ensemble framework, in which several classification algorithms are to be used, and a soft voting mechanism will be used to enhance the accuracy. Activity 3: Developed Resume Screening System, where the final system is implemented based on the trained ensemble framework.
3) Phase 3: Prototype Demonstration and Evaluation will evaluate the efficiency of the developed system. This stage will involve two tasks: Activity 1: Evaluation of Proposed System, during which the accuracy and efficiency of the system will be assessed on the basis of the performance metrics. Activity 2: Answer to Research Question, where findings are analyzed to determine how well the system addresses the research problem.

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A visual representation of the adapted methodology has been provided in figure 1.

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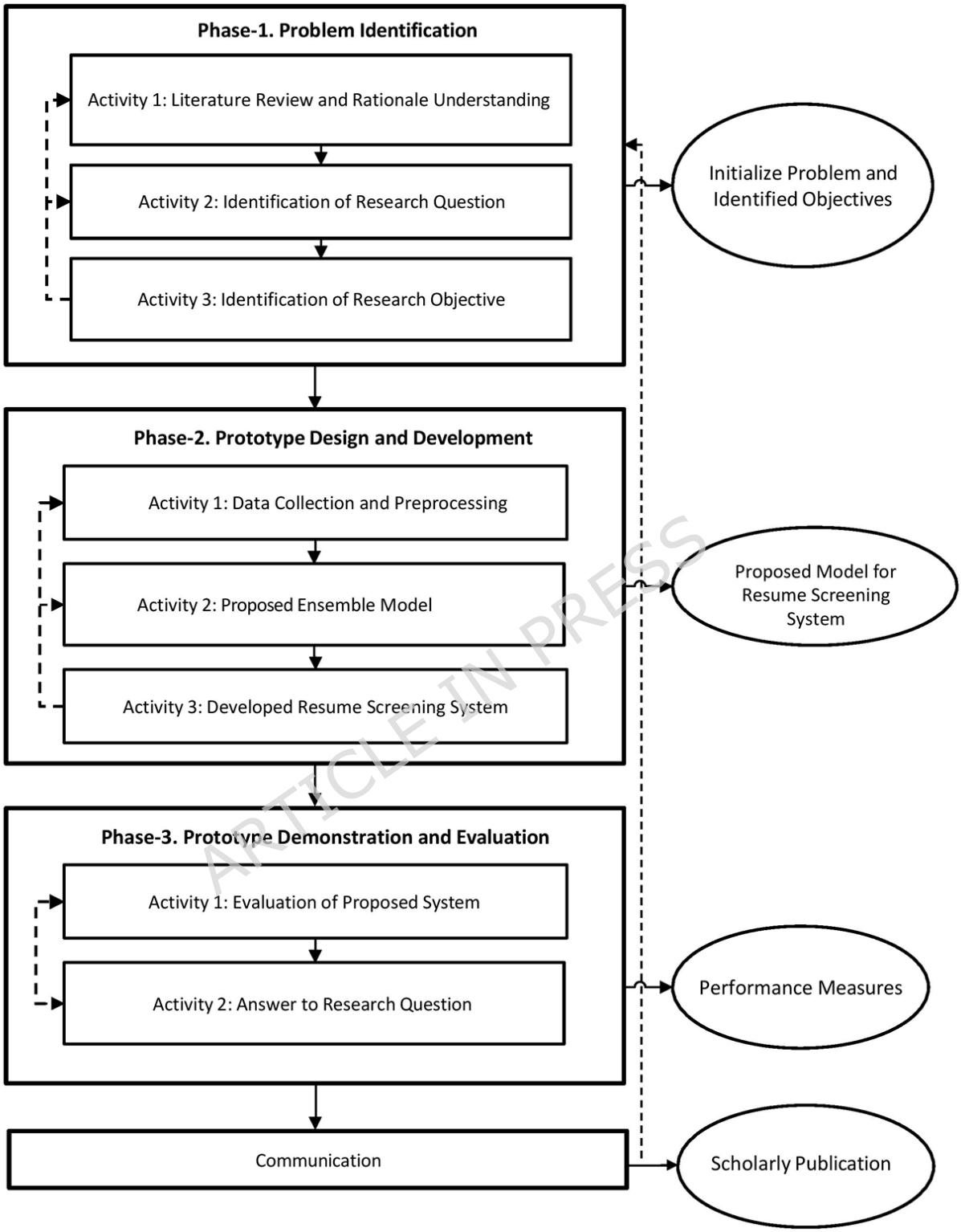


Figure 1. Design science research workflow for Resume Screening System

1. Problem Identification

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Problem Identification phase is concerned with the identification of research gaps, formulation of study objectives, as well as setting a clear direction of the research. It starts with a Literature Review, which analyzes the past research to define the limitations and explain why additional research is necessary. This results in the Identification of the Research Question which narrows the study and makes sure that the research covers gaps that remain unresolved. On this, the Research Objectives are established to provide a systematic channel of direction to the methodology and analysis. These aspects, covered in the Introduction and Literature Review sections, ensure the study is built on a solid theoretical and empirical foundation, paving the way for further research development.

2. Prototype Design and Development

Phase 2 focuses on data preprocessing, framework development, and implementation of the web-based resume screening system.

3.2.1. Data Collection

In this paper, we used the publicly available dataset of resumes available on Kaggle, which is a reputable site to access standardized datasets to conduct research on ML [15]. The dataset is a corpus of 962 resumes, which are organized into 25 different job roles, as shown in Figure 2. The dataset was specifically selected due to its format of raw resume text and respective job categories, which perfectly fits the primary goal of the given research project, which is to benchmark the performance and define the technical feasibility of numerous ML classifiers for automated text classification. The dataset is content-oriented, and it does not involve demographic or geographical metadata, but its adequate size and clear labeling offer a strong and objective base to conduct a thorough comparative study of

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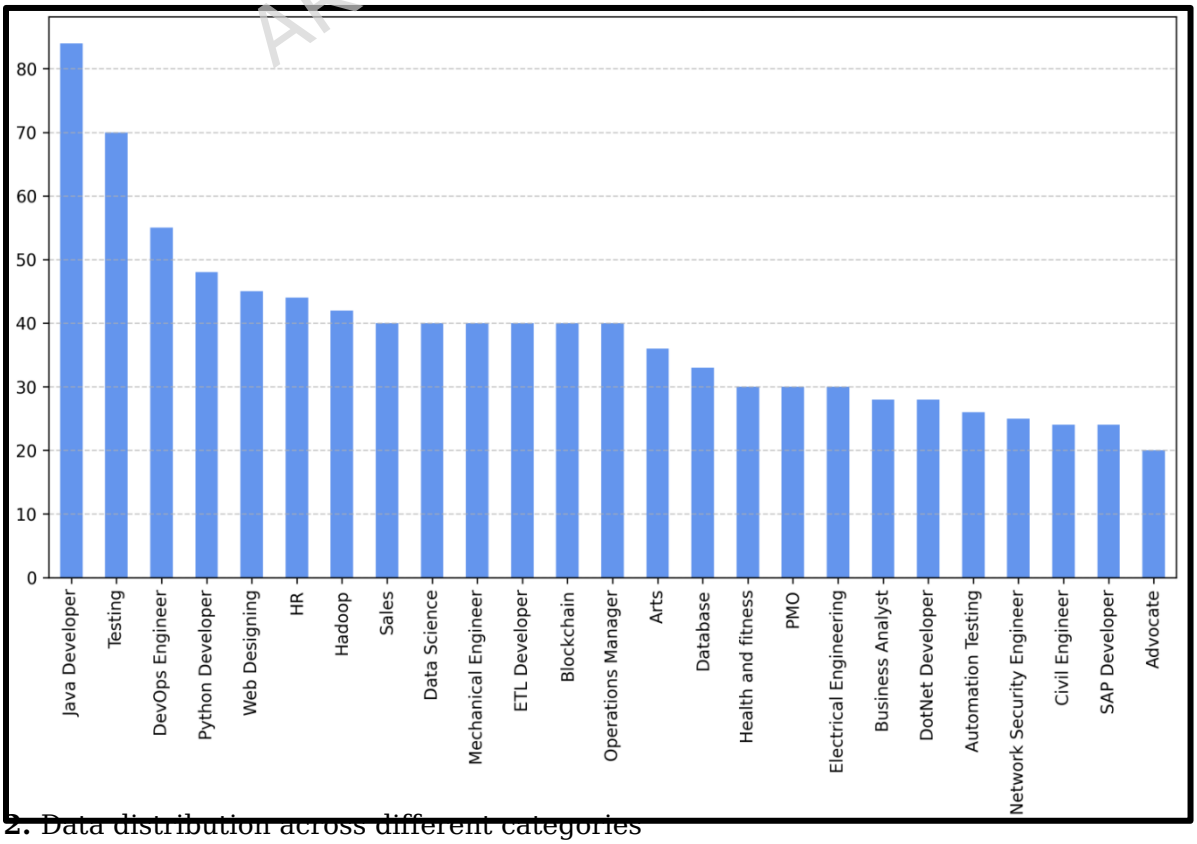


Figure 2. Data distribution across different categories

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algorithm accuracy. The raw data was formally accessed and subsequently passed to the preprocessing pipeline in the following sections.

3.2.2. Data Processing

Data preprocessing was performed to convert raw textual information into a form ready for ML model. The resultant subsections describe each process step by step.

- **Text Cleaning:** The text data in the Resume column underwent cleaning tasks to eliminate unneeded information along with data noise. A regular expression (re.sub) performed the task of removing special characters and numbers by replacing them with spaces. The cleaning process maintained only alphabetic characters since text analysis requires them. The text conversion to lower case normalization eliminated duplicate words from occurring because of case sensitivity rules (for example "Data" and "data" fused into one word).
- **Tokenization:** Tokenization broken-down text material into smaller fragments called tokens which normally include words phrases or symbols. The text analysis process starts with tokens which operate as basic components for performing further analysis. Tokenization divides text contents through distinct delimiters that consist of spaces together with punctuation marks as well as linguistic markers. The transformation of unstructured text into structure enables important findings to be extracted from original text sequence during this vital step. The sentence I love natural language processing can be broken down into five units, which are: [I, love, natural, language, processing]. The tokenization process utilizes different platforms that consist of libraries like Natural Language Toolkit (NLTK), spaCy and custom algorithms for its execution.The initial text preprocessing stage requires tokenization since it enables subsequent tasks including stopword removal, stemming, lemmatization and feature extraction in NLP.
- **Stopword Removal:** The list of stopwords includes common words which have minimal effect on text meaning (such as "the," "is," "and"). The NLTK library provided a predefined English stopwords list to remove unnecessary words from the document. The removal process diminished data complexity thus making analysis more efficient.
- **Lemmatization:** NLP employs lemmatization as a technique to transform words into their base or root forms known as lemmas by analyzing words within their context of meaning. Lemmatization provides better linguistic accuracy than stemming because it transforms word endings into valid language words through root formation. The word transformation process converts "running" into "run" and changes "better" to "good." The method detects the word's part of speech (POS) along with applying linguistic rules with WordNet as a reference. Through lemmatization multiple word forms belonging to the same lexical root united as a single term which enhanced the text data consistency. The NLTK library features its WordNet lemmatizer which was used to conduct the operation.

- **Feature extraction:** A TF-IDF vectorization technique enables the methodology to convert processed text data into numerical features for the feature extraction phase. Relative to the whole dataset, TF-IDF computes a statistical measure to determine word importance in documents. The method uses weight scales that are higher for specific document-focused terms which appear rarely throughout the entire dataset.

  The Term Frequency equation divides word frequency in a sentence by the complete sentence count.

$$\text{TF} = \frac{\textit{Number of times a word appears in a sentence}\ \text{no of occurrences of a word}}{\textit{Total number of words in the sentence}}\textit{total number of words in the sentence}$$

  The Inverse document frequency (IDF) calculation operates on a mathematical link between the log of total document sentences and the sentences that contain the queried word.

$$\text{IDF} = \log \frac{\text{Total number of sentences in the document}}{\text{Number of sentences that contain the word}}$$

$$\frac{\text{total sentences in the document}}{\text{sentences actually containing the word}}$$

  The multiplication of TF formula and IDF formula will give us a value in the form of a vector

$$\text{TF - IDF} = \text{TF} \times \text{IDF}$$

  The TF-IDF Vectorizer from the sklearn.feature_extraction.text module was used to convert the preprocessed text data into numerical features. The resulting feature matrix represents each document as a vector of TF-IDF values, where each value corresponds to the importance of a word in the document.

3.2.3. Machine Learning Classifiers

Following the preprocessing to identify the most effective base learners for automated resume screening, five diverse classification algorithms were benchmarked. These models were selected based on their varying mathematical approaches (linear, probabilistic, and tree-based) to ensure a comprehensive evaluation:

- **Logistic Regression:** Utilized as a baseline linear model to establish a performance floor using logistic function-based probability assessments.
- **Support Vector Machine:** Implemented with a linear kernel to identify the optimal hyperplane for high-dimensional text data separation.

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- **Naïve Bayes:** A Multinomial Naïve Bayes variant was employed, leveraging its strengths in handling discrete word frequencies and probabilistic dependencies.
- **Decision Tree:** A non-linear, tree-based classifier used to evaluate the effectiveness of recursive partitioning based on feature importance.
- **Random Forest:** An ensemble approach utilizing multiple decision trees to reduce variance and improve the stability of predictions.

3.2.4. Proposed Hybrid Ensemble Framework

This research proposed a Soft-Voting Hybrid Ensemble using the three best performing models, SVM, Random Forest, and Naive Bayes [16, 17]. These particular models have been chosen so as to provide architectural diversity with geometry (Support Vector Machine), tree-based (Random Forest), and probabilistic (Naïve Bayes) logic. Soft-Voting mechanism was selected instead of hard voting since it combines the predicted probabilities (confidence levels) of each classifier. This enables the ensemble to make a more fined-tuned choice by putting more emphasis on a model that is more certain about a particular category, hence lowering the error rate and reducing the robustness gap associated with individual models.

3.2.5. Implementation Settings and Hyperparameters

The following hyperparameter settings were used during training phase to guarantee reproducibility and reduce overfitting as show in Table 2. The settings in these environments were optimized by testing in order to make the model generalize to unseen resume data.

Table 2: Model Hyperparameters and Configuration Settings

Category	Parameter	Value	Rationale / Justification
TF-IDF Settings	ngram_range	(1, 2)	Captures both unigrams and bigrams (e.g., "Machine" and "Machine Learning").
	max_features	1000	Prevents high-dimensionality noise while retaining key descriptors.
Logistic Regression	max_iter	1000	Ensures convergence of the weight optimization process.
Decision Tree	max_depth	15	Limits tree growth to avoid memorizing training noise.
Random	n_estimators	100	Sufficient trees to stabilize variance and

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Forest			improve accuracy.
	max_depth	13	Enhances generalization by pruning deeper branches.
Linear SVM	C	0.1	Stronger regularization to encourage a wider margin and prevent overfitting.
	probability	True	Required for soft-voting probability aggregation.
Naïve Bayes	alpha	1.2	Higher Laplacian smoothing to handle rare words in unseen datasets.
Validation	test_size	30%	Stratified split to maintain class balance in evaluation.

3.2.6 Recommendation system

The system supports recommendation of resumes through an NLP processing step along with TF-IDF Vectorization that creates numerical features before KNN with cosine similarity identifies suitable matches. Cosine similarity calculates vector orientations to give precise recommendations through efficient matches. The system demonstrates excellent functionality when it relates job descriptions to suitable resumes.

Cosine Similarity

The measurement of vector similarity regardless of magnitude uses Cosine similarity as it's metric. The algorithm determines the cosine value between vectors that exist within multi-dimensional space with a measurement scale from -1 to 1. The numbers 1 for identical vectors and 0 for orthogonal (no similarity) vectors as well as -1 for diametrically opposed vectors represent the possible values of this measure. The formula for determining cosine similarity is presented as:

$$\text{Cosine similarity} = A \cdot B \div \|A\|\|B\|$$

The term A·B indicates the dot product between vectors A and B whereas ‖A‖ and ‖B‖ represent the vector magnitudes. Both job descriptions and resumes are converted into TF-IDF vectors because these numerical representations enable text data processing in the recommendation system. The algorithm uses cosine similarity for comparing the vector representation of the job description with each resume vector. Resume documents holding higher cosine similarity scores match job descriptions better since their contents deliver similar levels of harmony.

K-Nearest Neighbors Algorithm

As a simple non-parametric ML method K-Nearest Neighbors functions to classify and perform regressions on datasets. In this recommendation system, KNN is employed to find the most similar resumes to a given job description based on cosine similarity. The algorithm works by calculating the distance (or similarity) between the job description vector and all

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resume vectors in the dataset. It then selects the k nearest neighbors (resumes) with the smallest distances or highest similarity scores.

Algorithm

Input: Job_category, job_description and Resume description

Output: Best Resumes

Begin:

1. Get **Job_category**, **job_description** and **Resume description** from user.
2. Where **job_category** =? **job_description** =? and **Resume description** =?
3. Define **preprocess_text** function:
4. Remove non-alphabetic characters.
5. Convert text to lowercase.
6. **Tokenize** text.
7. Remove **stopwords.**
8. **Lemmatize** words.
9. Apply **preprocess_text** to "**Resume Description**" column to create "**Cleaned_Resume**".
10. Initialize **TfidfVectorizer**
11. Transform "**Cleaned_Resume**" into TF-IDFfeatures(X)
12. Extract target labels (y) from "**Job_category**" column.
13. Split dataset into training (70%) and testing (30%) sets.
14. Initialize models:
15. **Random Forest Classifier (rf)**
16. **Linear Support Vector Classifier (svm)**
17. **Multinomial Naive Bayes (nb).**
18. Create **VotingClassifier** ensemble with **rf, svm and nb**.
19. Train **ensemble model** on training data.
20. Predict labels for training and testing data.
21. Display classification results
22. Start **Resume Recommendation System**
23. Accept **job_description** from user.
24. Preprocess **job description**
25. Ensure job description has at least 5 meaningful words
26. Transform job description into **TF-IDF vector**.
27. Initialize **K-NN model** and fit on **TF-IDF** features.
28. Find nearest neighbors to job description vector.

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29. Filter resumes based on relevance threshold (0.7).
30. If relevant resumes found
31. Display top relevant resumes with cosine distances.
32. Else:
33. Notify user to refine job description.

End.

3.2.7 System development for Resume Screening

A functional prototype was created in form of a web-based system to illustrate the implementation of the proposed ensemble framework. The interface enables an HR professional or a hiring manager to fill in a comprehensive job description and add a set of resume submissions (CVs) in different formats (e.g., PDF, DOCX, TXT). When submitted, the textual content of both the job description and each CV is processed by the backend algorithm to provide a ranked list of the 5 most suitable candidates that would fit the job, based on their real-time similarity scores. Figure 3 shows the web-based user interface design of this prototype and Figure 4 demonstrates an input provided by a user.

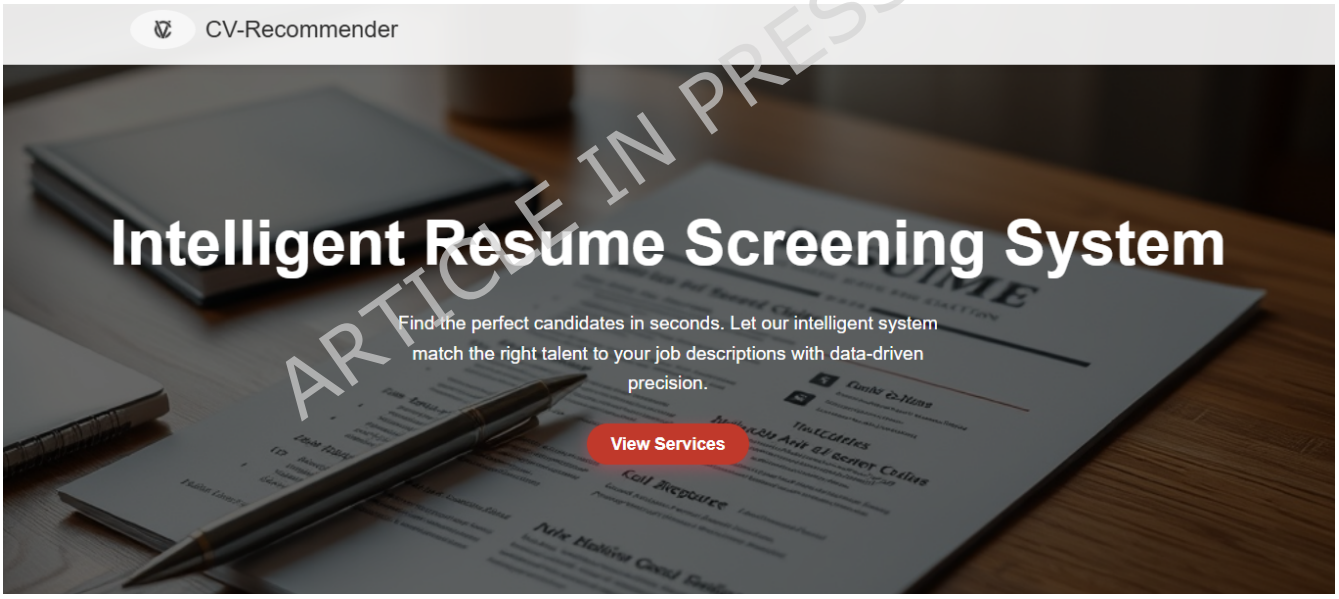


Figure 3. Web-based system for resume screening

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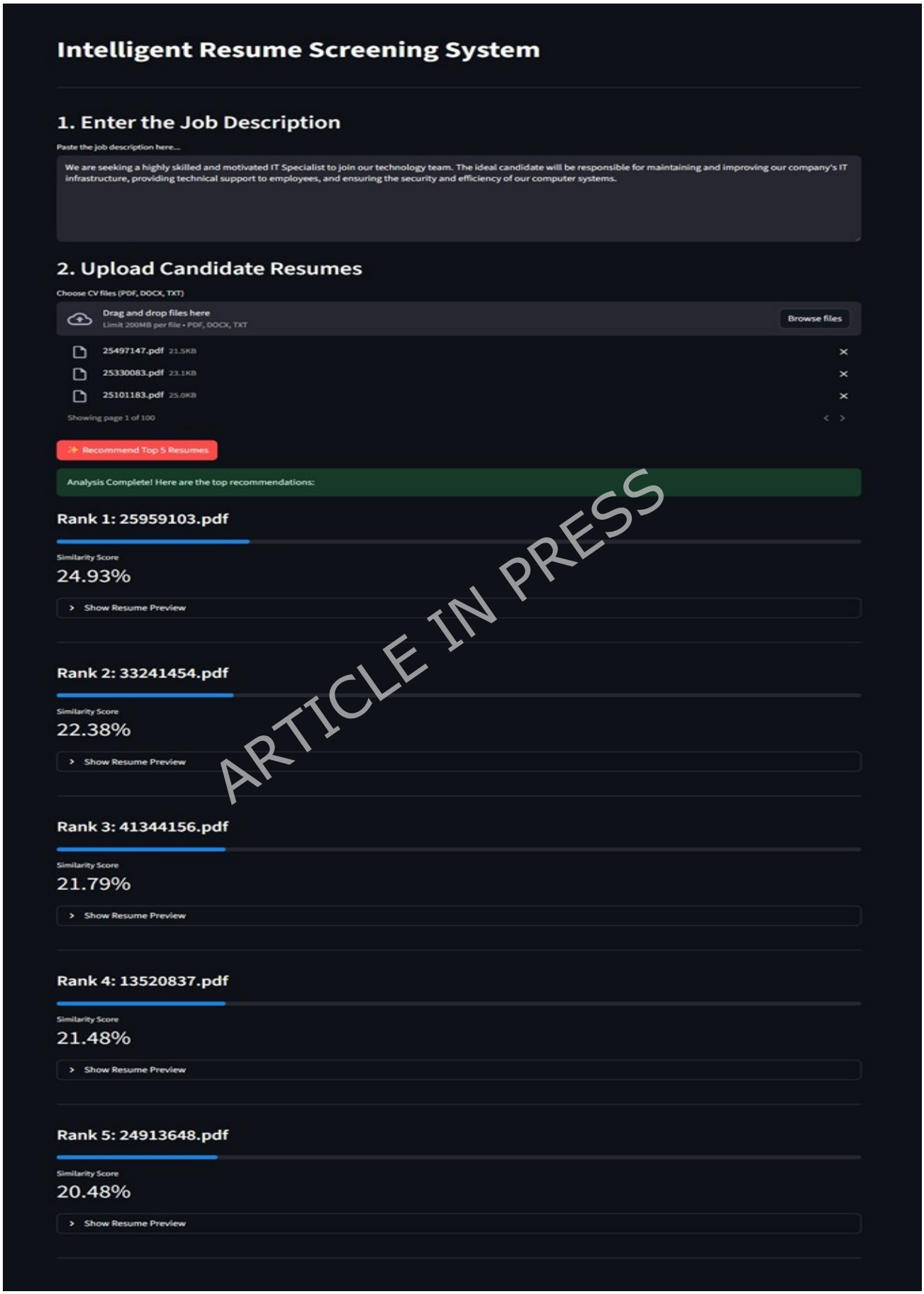


Figure 4. Recommendation of top 5 resumes

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3.3. Prototype Demonstration and Evaluation

This section evaluates the proposed framework and answer the research questions. The detail of this phase is given in section 4.

4. Results

A systematic, two stage experiment was conducted to measure the performance of the proposed framework. The initial step was dedicated to determining the baseline performance and the best model on the main benchmark dataset. The second step was aimed at strictly determining the strength and generalizability of the framework in a simulated real-world condition. In first phase to rigorously assess the performance of the classifiers, the following four metrics were used. These measures offer a more detailed view of the performance of a model than its accuracy, especially when screening resumes, where the cost of misclassification can be significant:

1) Accuracy is the proportion of all classifications that were correct, whether positive or negative. It is defined as:

$$\text{Accuracy} = \frac{\text{Number of correct classification instances}}{\text{Total number of instances}}$$

2) Precision is a measure of the model's reliability and is calculated as the ratio of true positives to all positive predictions.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive+False Positive}}$$

3) Recall measures how many of the actual positive cases were correctly identified by the model. It is important when missing a positive case (false negative) is more costly than false positives

$$\text{Recall} = \frac{\text{True Positive}}{\text{True positive+False Negative}}$$

4) F1-Score is the harmonic mean of precision and recall. It is useful when we need a balance between precision and recall as it combines both into a single number. A high F1 score means the model performs well on both metrics. Its range is [0,1].

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision+Recall}}$$

Initially, five different ML classifiers were evaluated for resume classification using TF-IDF feature extraction. As shown by Table 3 and figure 5, the performance of these individual models showed some major differences in their effectiveness. The lowest accuracies were obtained by the Decision Tree classifier and the Logistic Regression at 87.54% and 85.81% respectively. Although understandable, their performance was not optimal in a task where accuracy was important. Conversely, the Naive Bayes classifier was very reliable and had a high accuracy of 94.11%. The findings revealed three best-performing classifiers, namely Naïve Bayes (94.11%), Random Forest (97.23%), and especially Linear SVM, which became the most successful single model with an overall highest accuracy of 97.57%. These three models were chosen

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strategically as a foundation of a hybrid ensemble framework because they performed well individually and, above all, they have different algorithmic methods (probabilistic, tree-based and geometric).

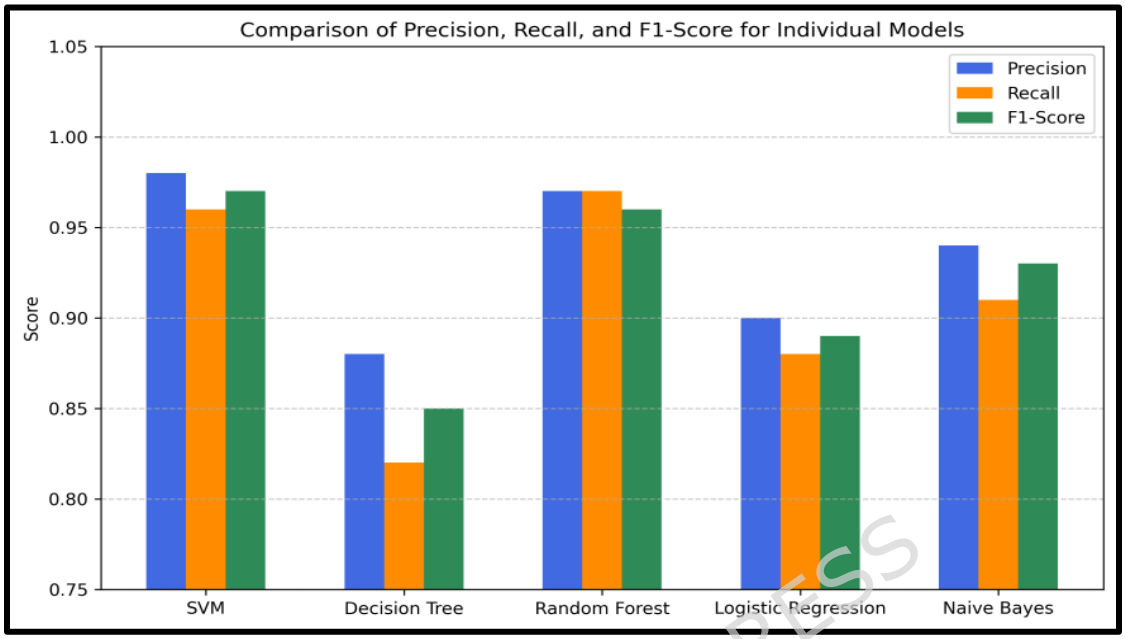


Figure 5. Comparison of Individual Models

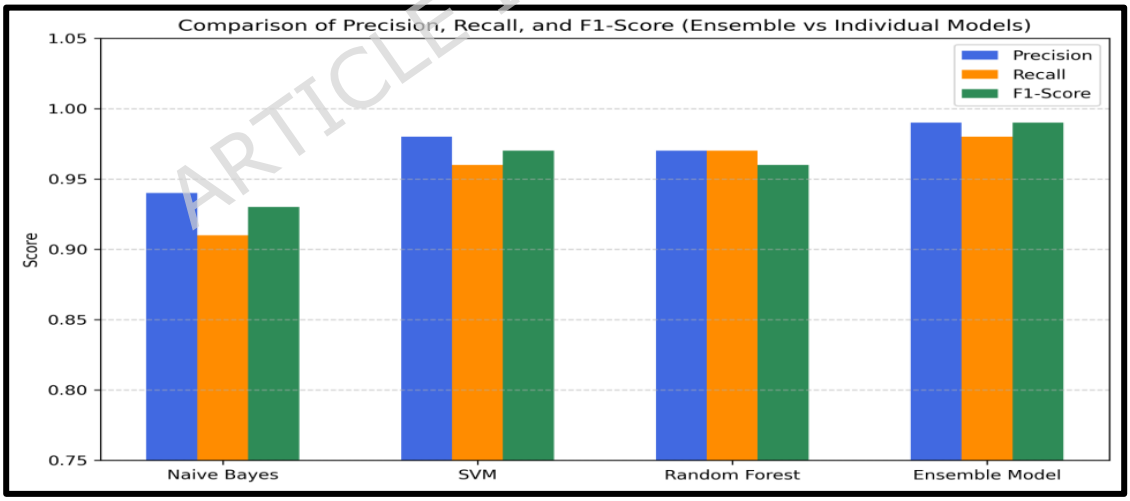


Figure 6. Comparison of Ensemble Model with other Models

To further enhance accuracy and reliability, a soft-voting ensemble classifier was constructed using these top three models. The proposed ensemble framework, as presented in Table 3, does not only obtain the best accuracy but also have better Precision (0.99) and Recall (0.98). Although the single models such as SVM and Random Forest perform well, the ensemble framework manages to balance the trade-off between precision and recall

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effectively leading to an almost perfect F1-Score (0.99). This overall analysis helps to overcome the weakness of using accuracy alone and ensures that the model is effective in reducing the number of false positives and false negatives in the screening of resumes.

4.1 Analysis of Confusion Matrix

Although aggregate accuracy scores give a broad summary of the performance of a model, it is necessary to conduct a more detailed analysis of the confusion matrices to identify the underlying classification behaviors and residual errors of a particular model. The Naive Bayes confusion matrix, with a respectable overall accuracy (approximately 94.5% as seen in Figure 7) suffers from the inability to deal with categories with high lexical overlap, e.g., between Advocate and Automation Testing, which is a consequence of depending on the probability of words outside of the context. The Random Forest classifier (~98.6%) corrects many of these errors but displays its own subtle weakness: minor confusion with generalist roles like 'DevOps Engineer' and 'HR', where profiles contain a wide but inconsistent set of keywords, shown in Figure 8.

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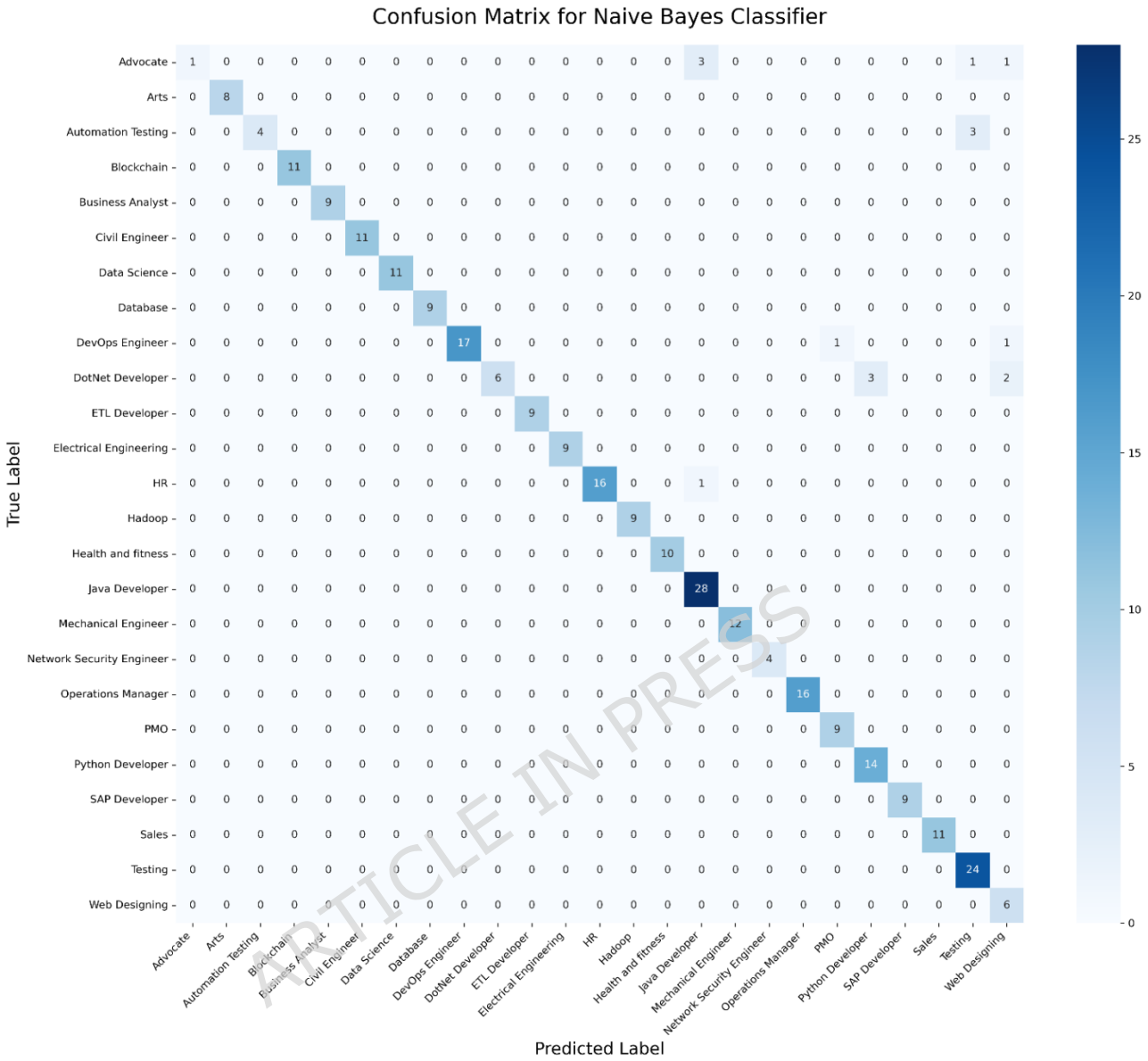


Figure 7. Confusion Matrix for naive bayes

The Linear SVM, which is the best-performing individual model (~99.0%), shows the cleanest matrix among the three, but its limited number of errors indicates a defining weakness: slight confusions between highly abstract roles such as Data Science and Business Analyst where the decision boundary itself is non-linear, as seen in figure 9. The confusion matrix of the ensemble framework that is the most precise is displayed in figure 10 and presents the real value of the proposed system. It proves to be remarkably able to address the particular, independent weaknesses of its component models: where Naive Bayes was baffled by lexical overlap, its errors were outvoted by the other models; where Random Forest was vague, the high confidence of SVM was used to give a corrective signal; and where SVM was confused by non-linear boundaries, the combined probabilistic and tree-based reasoning helped resolve the classification. The soft-voting mechanism successfully averages

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out these individual model deficiencies, resulting in a system with not just the highest accuracy, but also the most robust and reliable classification behavior across all 25 categories, empirically validating the superiority of the hybrid approach.

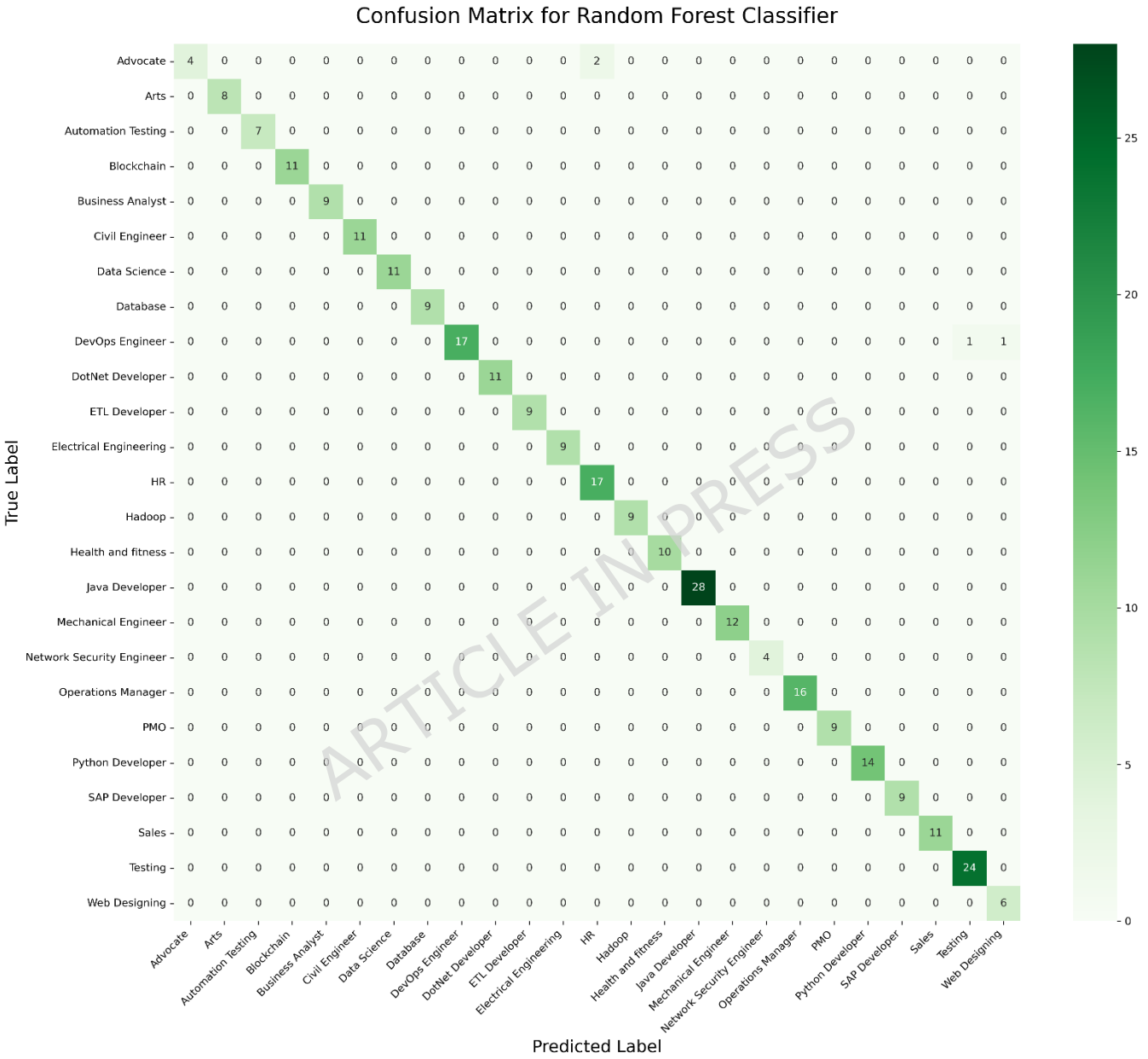


Figure 8. Confusion Matrix for random forest

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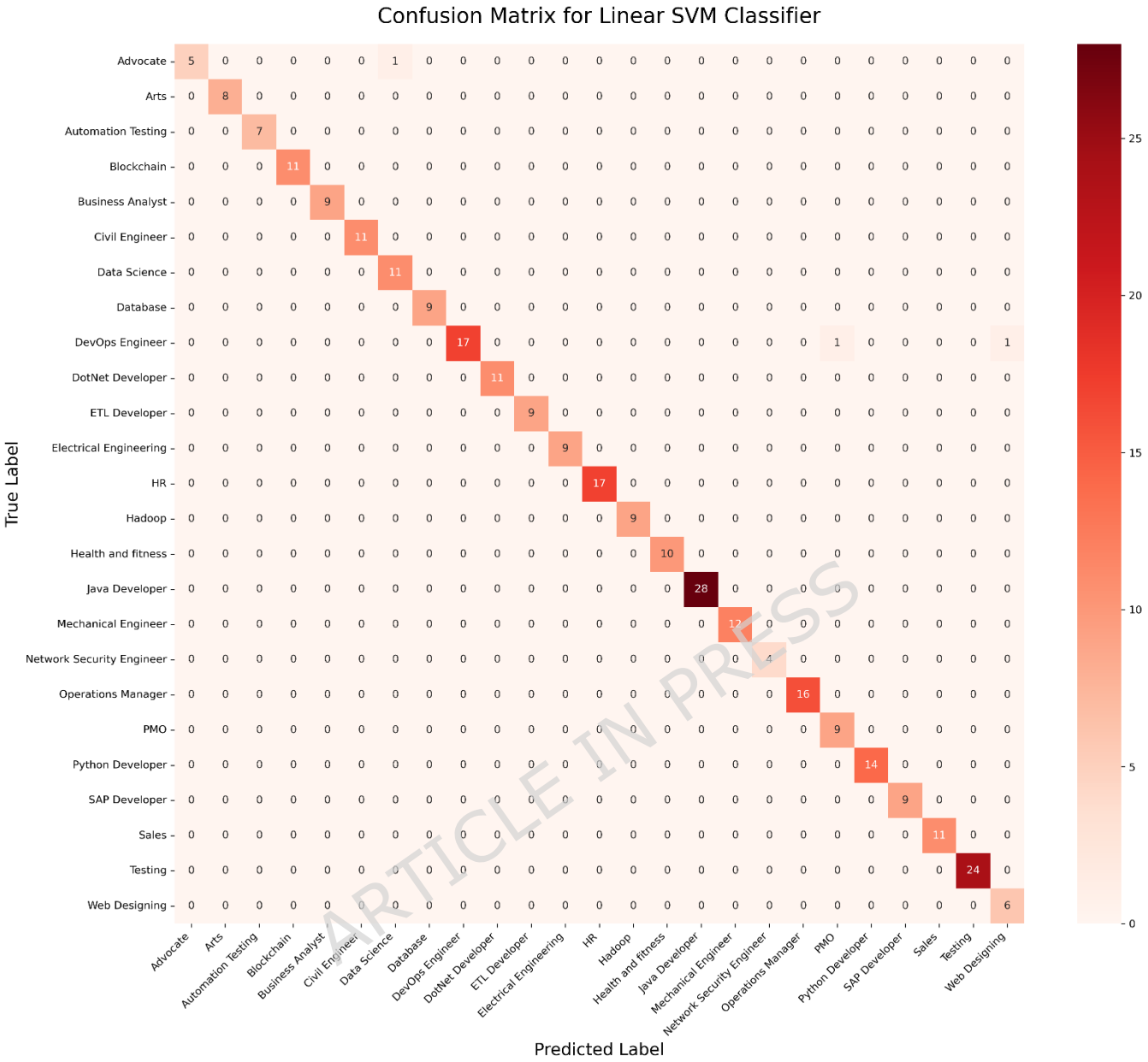


Figure 9. Confusion matrix for SVM

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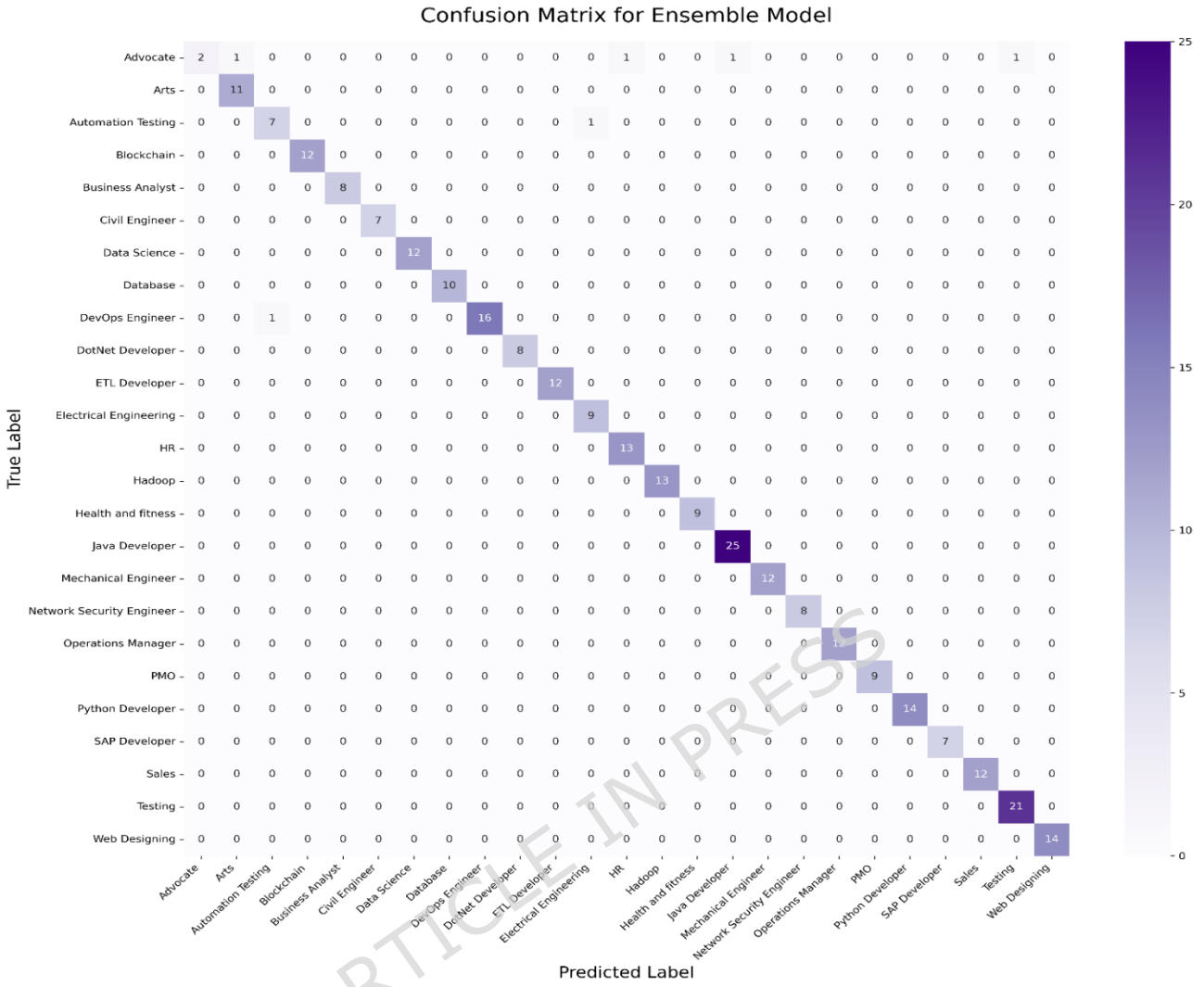


Figure 10. Confusion Matrix for ensemble Frameworj

4.2. Robustness and Generalizability Test

In order to strictly test the applicability of the proposed framework a secondary validation corpus of 598 resumes was systematically curated in a multi stage process. Resumes were specifically pooled across different online sources, including web searches targeting publicly indexed profiles, subsets of irrelevant smaller datasets on Kaggle, and other career portal templates to make the greatest structural diversity. In order to guarantee a consistent preprocessing pipeline and remove extraction errors, all the raw files in this collection were transformed into a standardized PDF format, followed by text extraction. Ground-truth labelling of this dataset was done in a manual blind-reviewing process by two HR domain experts who had more than five years of recruitment experience. These professionals manually screened the standardized PDFs to classify each resume, and the labels were correct and did not have the automated labeling noise that is common in benchmark datasets.

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Due to such an expert-directed curation, the resulting validation corpus is a very heterogeneous sample of professional backgrounds. As shown in Figure 11, the dataset consists of seven different professional groups: Information Technology (IT), Finance, Human Resources (HR), Business Development, Designer, Teacher, and Sales. This professional diversity, represented in the category-wise percentage distribution of Figure 11, makes sure that the benchmark of validation is not biased to one domain and very close to the reality of talent acquisition which is unpredictable. This conscious combination of formatting styles and the use of keywords in various industries presents a difficult and unforgiving landscape to the classifiers. The top-performing individual classifiers and the proposed Ensemble framework were applied to this unseen dataset in a 'zero-shot' manner, without any retraining or fine-tuning. The relative findings of this robustness test, provided in Table 4, reveal the evident dominance of the hybrid ensemble framework. Whereas standalone models suffered greatly when exposed to this type of heterogeneous data, namely, the Linear SVM accuracy declined by 4.42 percent and Random Forest by 6.21 percent, the proposed Ensemble framework retained an impressive accuracy of 97.34 percent. This represents a negligible performance drop of only 1.10% from its benchmark score. These empirically validated results on a diverse and expert-labeled dataset show that the ensemble framework is not only highly accurate but also incredibly stable and reliable to be used in the real-world HR setting.

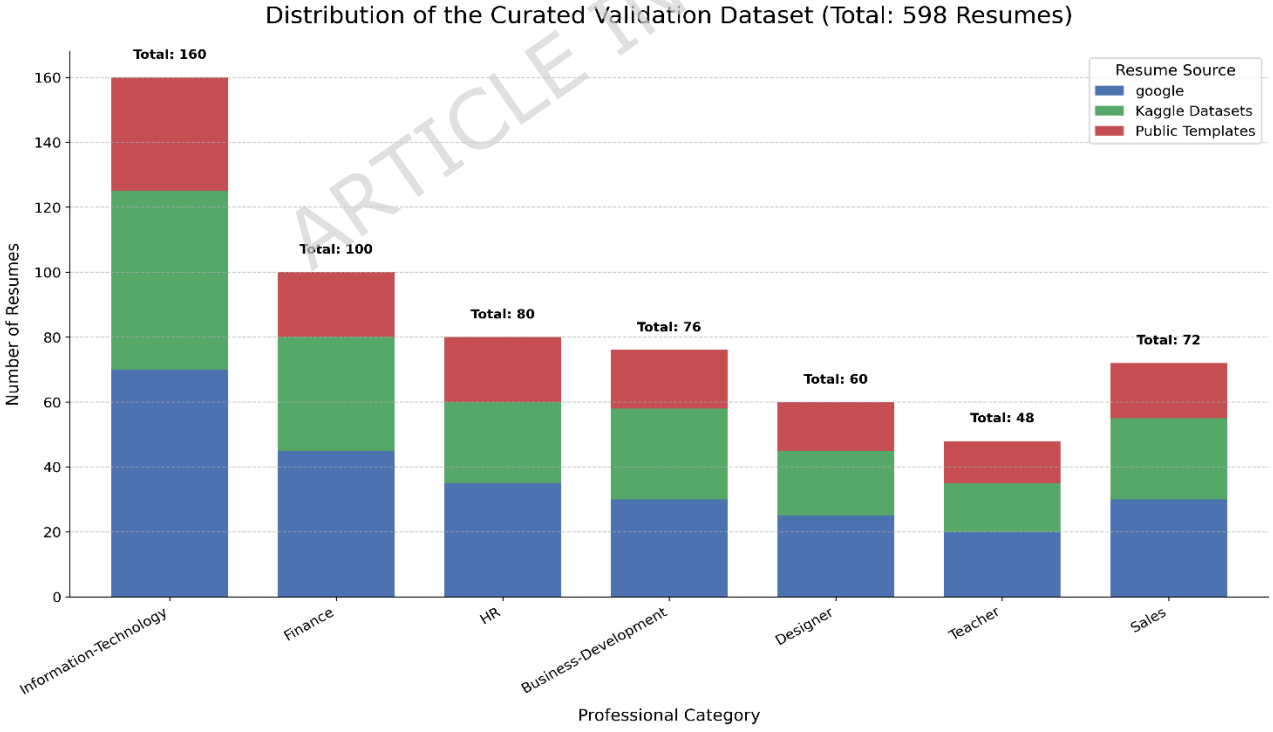


Figure 11. Distribution of the Curated Validation Dataset

Table 3. Results of Models

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Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	85.81	0.90	0.88	0.89
Decision Tree	87.54	0.88	0.82	0.85
Naive Bayes	94.11	0.94	0.91	0.93
Random Forest	97.23	0.97	0.97	0.96
Linear SVM	97.57	0.98	0.96	0.97
Proposed Ensemble Framework	**98.44**	**0.99**	**0.98**	**0.99**

Table 4. Comparison of Model Performance on Original Test Set vs. Unseen Generalization Dataset

Model	Accuracy on Original Test Set	Accuracy on Unseen Dataset	Performance Drop
Logistic Regression	85.81%	80.05%	-5.76
Decision Tree	87.54%	79.94%	-7.61
Naïve Bayes	94.11%	87.48%	-6.63%
Random Forest	97.23%	91.02%	-6.21%
Linear SVM	97.57%	93.15%	-4.42%
Ensemble Framework	98.44%	97.34%	-1.10%

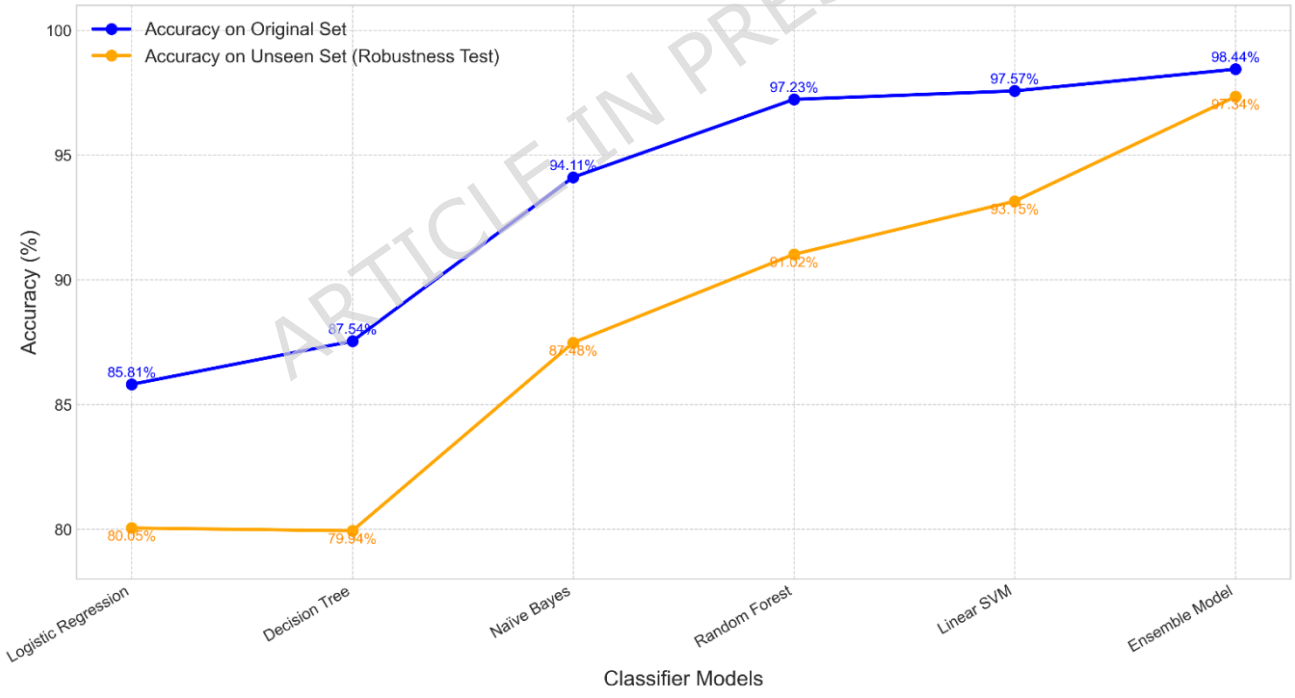


Figure 12. Model Performance Degradation: Original vs Unseen Dataset

4.3. Post-Hoc Interpretability and Empirical Fairness Audit

To overcome the conventional black-box implementation and to meet the

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paramount need of transparency of algorithms, a post-hoc interpretability audit was conducted on the trained hybrid ensemble framework. The analysis is critical where the predictive performance of a framework is crucial as much as the internal rationale of the model. The analysis of the best predictive weights generated by the TF-IDF feature matrix reveals the particular semantic anchors that generate the classification process in different professional fields. This audit enables a finer evaluation of whether the framework has acquired the proper professional differences or is accidentally depending on the noise that is non-relevant, thus giving a stringent base to the reliability of the system. The feature importance analysis results, as shown in Table 5, give a clear picture of the semantic reasoning of the framework. The top 15-feature ranking shows that the predictive weights and domain specific competencies are closely aligned, and such cross domain terms like Experience, Management, and Project are good indicators of professional seniority. More to the point, extremely specialized tokens such as python, Java and SQL tokens relating to Information Technology jobs or Financial and Analysis tokens relating to Finance jobs carry a lot of weight. This validates the fact that the ensemble framework has been able to determine the linguistic boundaries of the various industries in a successful manner, which guarantees the uniform and logical classifications in the presence of the heterogeneous data structure in secondary validation dataset.

Table 5: Top 15 Features (Keywords) by Predictive Weight and Semantic Association

Rank	Feature (Token)	Global Importance	Category Association	Rationale for Weighting
1	Experience	0.085	Cross-Domain	Core indicator of professional seniority.
2	Management	0.072	HR / Business	High frequency in leadership & admin roles.
3	Python	0.068	IT / Data Science	Technical discriminator for specialized roles.
4	Java	0.065	Software Dev	High-precision marker for backend IT roles.
5	SQL	0.061	Data / Database	Essential skill for analytical & IT domains.
6	Project	0.058	Cross-Domain	Indicates task-oriented professional history.
7	Sales	0.055	Marketing / Sales	Highly specific keyword for revenue roles.

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8	Financial	0.052	Finance / Banking	Semantic anchor for accounting & audits.
9	Development	0.049	IT / Engineering	Broad discriminator for technical functions.
10	Customer	0.045	Sales / Support	High relevance in service-oriented roles.
11	Testing	0.042	IT / Quality	Niche marker for QA and Automation roles.
12	Analysis	0.040	Finance / Data	Key descriptor for research & logic roles.
13	Recruitment	0.038	HR	High-weight token for administrative HR.
14	Design	0.035	Designer / Creative	Distinctive keyword for creative professions.
15	Team	0.032	Cross-Domain	Soft-skill indicator of collaboration.

This feature audit is the empirical basis of the study claims that algorithms are fair and make choices based on merit. The feature analysis of the most weighted features is a confirmation that the system is purely dependent on technical capabilities, professional functions, and industry specific terminologies to make its predictions. Importantly, a careful examination of the best predictive features showed that there will be no demographic identifiers or proxy variables at all, including tokens connected to gender, age, or ethnicity, making sure that the screening process will not be affected by subjective human biases. The hybrid ensemble framework successfully addresses ethical issues associated with discriminative input processes by converting lemmatized TF-IDF vectors, prioritizing skill based markers, and establishing its usefulness in practical talent acquisition in real-world recruitment contexts.

4.4 Statistical Analysis

To offer a strict analysis of the framework performance, we have reported the average accuracy and standard deviation (error margins) with 5-fold stratified cross-validation as shown in Table 6. The ensemble framework had a mean accuracy of 97.47%±1.00. The statistical significance was confirmed through Paired T-tests. Although the ensemble had similar performance to Random Forest and Logistic Regression on the primary benchmark dataset ($p>0.05$), it had statistically significant better performance compared to Naive Bayes, Decision Tree, and Linear SVM. Importantly, the hybrid ensemble framework has been selected due to its

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stability with different data distributions. Although individual models such as Random Forest are incredibly efficient on the controlled benchmark data, they tend to increase the variance when tested on unseen, heterogeneous data. The ensemble framework, combines geometric, probabilistic and tree-based logic, offers a stronger decision boundary leading to the improved generalizability seen in the robustness tests.

Table 6: Comparative Performance and Statistical Significance (5-Fold Cross-Validation)

Classifier	**Mean Accuracy ±SD**	**p-value (Relative to Ensemble)**	**Result**
Hybrid Ensemble	**97.47%±1.00%**	-	-
Random Forest	97.77%±1.33%	0.47780	Comparable
Naive Bayes	92.28%±1.44%	0.0001	Significant
Linear SVM	97.57%±2.14%	0.0000	Significant

4.5 Comparison with State-of-art

As seen in the comparative analysis in Table 7, although the existing literature has set a high standard of classification accuracy, it also shows that there are vulnerable gaps in it which are filled by the current study. Particularly interesting is the work of Ali et al. [7], who had a near-perfect accuracy of 99.6% on the same 962-resume dataset as the current study, which shows that well-tuned classical models can be very effective indeed. This movement towards more complex architectures can also be observed in the publications by Al-Quhfa et al. [14] and Shen et al. [9] who investigated powerful ensemble designs and state-of-the-art deep learning models such as BERT. Nonetheless, this push to complexity creates a critical performance paradox, as seen with Kuppala et al. [10] with their CNN+GRU model, which when using the same exact dataset, performed disastrously with an accuracy of only 88.0, which is far behind simpler classical methods. With these diverse methods, however, the table reveals a key and constant constraint in all these state-of-the-art studies: a critical robustness gap. Although high accuracy is repeatedly obtained, they are only reported on test splits of the original source data, and their suitability in practice has not been demonstrated. This is the major gap that the current study is intended to address. Interestingly, the work is the only one of the compared studies that empirically validates a newly curated, heterogeneous dataset, directly filling in the robustness gap in the existing literature.

Table 7. State-of-the-Art Comparison

Id	**Methodology**	**Dataset Used**	**Reported Result**	**Validated on Unseen**

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				Data?
[1]	Baseline Classical ML (Linear SVM, RF, NB)	Kaggle Dataset	78.53%	No
[7]	Optimized Classical ML: 9 classifiers (LSVC best)	962 Resumes	99.6%	No
[9]	State-of-the-Art Deep Learning (BERT + GCN)	Private Dataset	93.4%	No
[10]	Deep Learning Paradox (CNN+GRU Ensemble)	962 Resumes	88.0%	No
[14]	Broad ML Comparison: 9 classifiers (Random Forest best)	Private Yemeni Recruitment Data	92.8%	No
Proposed Framework	Robust and Efficient Hybrid Ensemble	962 Resumes	98.44%	YES

4.6 Answers to Research Questions

The experimental findings from this study provide clear answers to the research questions:

To what extent can a hybrid ensemble framework improve upon the benchmark accuracy of high-performing individual classifiers in the context of automated resume screening?

The findings indicate that the hybrid ensemble framework provides a significant and quantifiable enhancement even in comparison to the standalone classifiers that perform the best. As the first performance assessment described in Section 4 indicated, the ensemble framework attained a final accuracy of 98.44, higher than the highest accuracy of the best-performing single model, Linear SVM, which is 97.57. This demonstrates that the architectural benefit of integrating varied classifiers results in a more generalized decision boundary and statistically significant decrease in misclassifications, therefore creating a new, higher precision standard on this task.

Beyond standard accuracy metrics, how does the real-world robustness of a hybrid ensemble framework compare to that of standalone classifiers when evaluated on a diverse, unseen, and heterogeneous dataset?

The comparison demonstrates a sharp and striking variation in real-world strength, which is well-documented and confirmed in the robustness validation test. The standalone classifiers, although with high benchmark scores, were brittle with a performance deterioration of -4.42% to -7.61% on the unseen dataset. Conversely, the hybrid ensemble framework was remarkably stable, with only a small performance decrease of -1.10. This gives the conclusive

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empirical data that the ensemble architecture is not simply more precise in theory, but far more important, the metric of reliability in real-life use and hence actually deployable and reliable system.

5) Conclusion

This paper introduces a sophisticated ML-based automated resume screening framework that is capable of increasing the efficiency of recruitment through objective candidate classification. The study had a sequential implementation process, which started with the benchmarking of five separate classifiers. These results were analyzed and the best performing models were selected and combined to form a refined hybrid ensemble framework. The ensemble, which used a Soft-Voting mechanism to integrate geometric, probabilistic, and tree-based logic, had a better peak accuracy of 98.44%. Statistical validation with Paired T-tests show that the superiority of the ensemble in comparison with the individual configurations is statistically significant and scientifically valid. One of the most remarkable contributions of this work is the empirical validation of model robustness on an entirely invisible, expert-labeled heterogeneous dataset, in which the ensemble was exceptionally stable with only a 1.10% decrease in performance. Moreover, a presence of an interpretability audit will offer such verifiable evidence that the system is merit-centric when making decisions and not biased in demographics. This demonstration of a working web-based system also reflects the practical feasibility of the research, revealing how the high accuracy ensemble framework can be effectively implemented in practice in recruitment processes. Future research will include incorporating formal measures of fairness and improved explainable capabilities to promote more trust and transparency in automated recruitment contexts.

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Author contributions: M.H. conceptualized the study, designed the methodology, and supervised the overall research process. M.H. and F.H. conducted the experiments, including data preprocessing, model development, and algorithm implementation. M.H., F.H., and A.S.A. performed the data analysis and statistical evaluations. M.H., A.S.A. and R.G contributed to the interpretation of the results and the preparation of visualizations. M.H. and F.H. drafted the manuscript, while A.S.A. and R.G. critically revised the manuscript to enhance its content and presentation. All authors reviewed and approved the final version of the manuscript.

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Declarations competing interests: The authors declare no competing interests.

Data availability:

The dataset used for model development is available at: https://www.kaggle.com/datasets/jillanisofttech/updated-resume-dataset

The dataset used for validation of the proposed system is available at: https://github.com/drmhamid/Automated-Resume-Screening-Using-Machine-Learning

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